

Full Name: NEMO

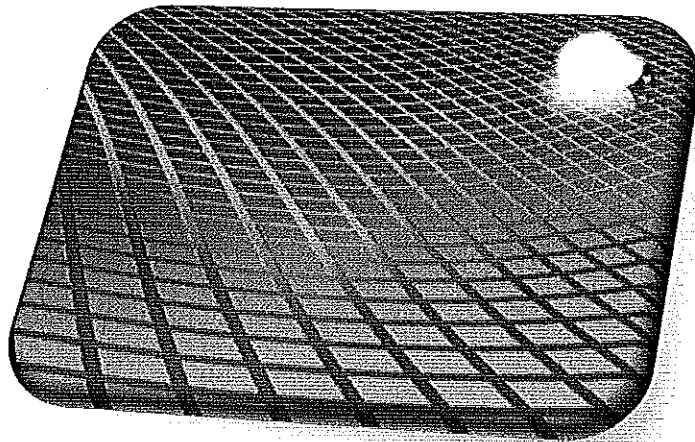


Ellenbrook
CHRISTIAN COLLEGE

**MATHEMATICS
YEAR 8**

Problem 2 – Counting Squares

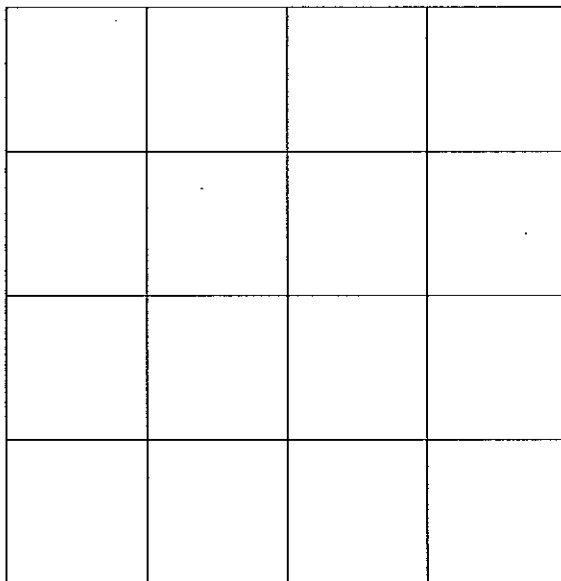
Semester 1 2014



Information for Students

Working time:	20 minutes
Marks available:	10 marks

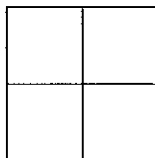
Consider the following 4×4 square grid. How many squares can you see? Think about this carefully.



The number of squares is quite difficult to determine (there are more than 16).

Answer the following questions to help you with the solution to the above problem.

1. Consider the 2×2 square grid.



- a) How many 1×1 squares are there?

[1]

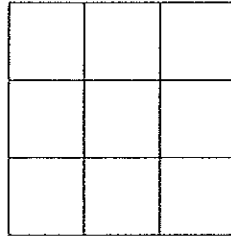
4

- b) How many 2×2 squares are there?

[1]

1

2. Consider the 3x3 grid.



a) How many 1x1 squares are there?

[1]

9

b) How many 2x2 squares are there?

[1]

4

c) How many 3x3 squares are there?

[1]

1

d) How many squares are there in total in this 3x3 grid?

[1]

14

3. a) By carefully looking at the number pattern developed in the previous two cases of a 2×2 and a 3×3 grid, determine how many squares are in the original 4×4 grid.
(Show clearly how you used this number pattern.) [2]

16 4×4

9 3×3

4 2×2

1 1×1

Total 30

- b) Determine how many squares there are in a 10×10 grid. [2]

100

81

64

49

36

25

16

9

4

1

Total 385

End of Problem